A constrained detector, evaluated end to end

The research question is when added representation and computation improve useful detection.

01 URL-only structural model → selective character transformer

02 Monitoring, operating-point risk and real HTTP execution

03 Revision: 12 October 2026 · Krti Tallam

Manuscript title; §§1.3–1.5

Three questions, unchanged numerical commitments

Representation, monitoring and inline operation must be judged together.

01 RQ1: incremental recall with internal and external FPR ≤1%

02 RQ2: shift sensitivity ≥80%, reference alerts ≤5%, useful future routing

03 RQ3: recall loss ≤2 points, calls ≤30%, p95 ≤200 ms, errors <0.1%

§§1.4.1–1.4.3; corrected-primary-gates.csv

Why each comparison exists

Each intervention answers a different mechanism question.

01 Original: 22 checks and 25 operational groups

02 D / F: refitted invariant representation / frozen-model scheme spelling

03 R / C / P / S: routing / actual computation / fit audit / HTTP client design

§§3.15–3.16; Appendices A–C

Structural information adds recall; specificity does not transfer

RQ1 separates representation value from the chosen operating point.

01 Structural-over-length gain: +64.07 points internal; +76.81 external

02 Internal structural FPR: 0.7747%; external certified FPR: 90.5487%

03 Original gate selects no evaluation calls, so cascade recall gain is zero

§§3.8, 4.2; Tables 4.2–4.3

Diagnose the shared errors before redesigning the gate

A selector cannot fix errors shared by both unchanged decisions.

01 Original certified negatives: 2,261 shared false positives / 2,497

02 Transformer repairs 0 and introduces 218 negative errors

03 Observed unchanged-decision selector floor: 90.5487% FPR

§4.8.1; saved-score-routing-replay-v2.json

Changing the scheme moves both classes

F isolates spelling sensitivity with model, scaler and threshold fixed.

01 Retained spelling: TP 69/69; FP 2,261/2,497

02 Forced HTTPS: TP 22/69; FP 50/2,497

03 2,211 fewer false positives, but 47 fewer positive detections

§4.10.1; Table 4.21; F-formatting.csv

A wider gate must select repairs, not simply more cases

Development benefit did not transfer to R's retained domain-balanced frame.

01 R: 1,392 positive + 1,262 negative domains; one URL per domain

02 Symmetric gate: 21 calls / 2,654; 0 selected repairs; 18 new false positives

03 All systems detect 1,392 positives; structural FP 1,028, symmetric FP 1,046

§§3.16.1–3.16.2, 4.10.2; Table 4.22; R-routing.csv

More precise recall evidence, without changing the original verdict

Agreement must be accompanied by a nondegenerate uncertainty interval.

01 Original 69 positive pairs: corrected 95% envelope ±6.1533 points

02 R's 1,392 pairs: 95% envelope ±0.3143 points

03 Two new recall-only noninferiority statements pass the −2-point margin

§§3.8, 4.10.2; Table 4.3; R-paired-contrasts.csv

Rare expert calls appear in p99, not necessarily p95

C measures actual model work, separate from the HTTP service.

01 Symmetric: 17 expert calls / 1,000 requests; 1.646× structural time

02 Median run p95: 0.223 ms; p99: 6.769 ms

03 Always-transformer: 30.781× structural summed-scoring time

§4.10.3; Table 4.23; C-cost-summary.csv and C-all-arms.csv

A monitor signal is not automatically a useful action

RQ2 evaluates both distribution recognition and what happens afterward.

01 External window detection: 115/132 = 87.12%

02 Reference false alerts: 28/252 = 11.11%

03 Future routing adds 218 false positives and no gold detections

§§4.3, 5.1; Tables 4.4–4.5

The client-path modification has a measured benefit

S changes client connection ownership while keeping structural scoring fixed.

01 Median paired p95 reduction: 79.68% at concurrency 64

02 Worker successful p95: 71.999 ms; 0 errors / 100,000 attempts

03 99,999 / 99,999 comparable predictions agree

§4.9; Tables 4.18–4.20 and B.1

Verify fitting inputs instead of trusting a plausible metric

P directly links each new shuffled-label fit to its consumed label vector.

01 Five seeds, 42–46; all converge below 5,000 iterations

02 Validation AUC range 0.3940–0.6965; AP 0.4027–0.7683

03 Every outcome retained; no seed selection or favorable reruns

§4.10.4; Table 4.24; P-all-seeds.csv

What the dissertation adds to established methods

The contribution is tested engineering knowledge about constrained decisions.

01 Measure same-example repair before optimizing deferral

02 Test low-FPR discrimination alongside representation invariance

03 Separate model calls, actual computation and client-observed service cost

§§2.2–2.9, 5.2; Tables 2.1 and 5.1

Hypothesis decisions and positive findings are different levels

All original questions have direct answers; every numerical requirement remains visible.

01 Original checks: H1 5/10, H2 1/4, H3 2/8 pass

02 Joint H1–H3: unsupported; all constituent conditions are required

03 Positive findings: structural gain, exact invariance, recall-only NI and HTTP improvement

§§4.7, 5.1–5.6; Tables 4.14 and 5.2; corrected-primary-gates.csv

The technical answer is actionable

Change what the evidence identifies, and test every consequence.

01 Shared decision errors require changed information or operating-point development

02 A useful route must find repairs while controlling harms and compute

03 Optimize the measured service path without calling it better detection

§§5.2–5.6

A traceable review package

A reader can move from question to method, table and retained evidence.

01 Manuscript + seven-comment response + extended methods

02 Corrected ledger + F/R/C/P aggregates + freeze identities

03 Original record preserved; new record-level identities remain private

Appendices A–C; EVIDENCE_MAP.txt; VERIFICATION.txt